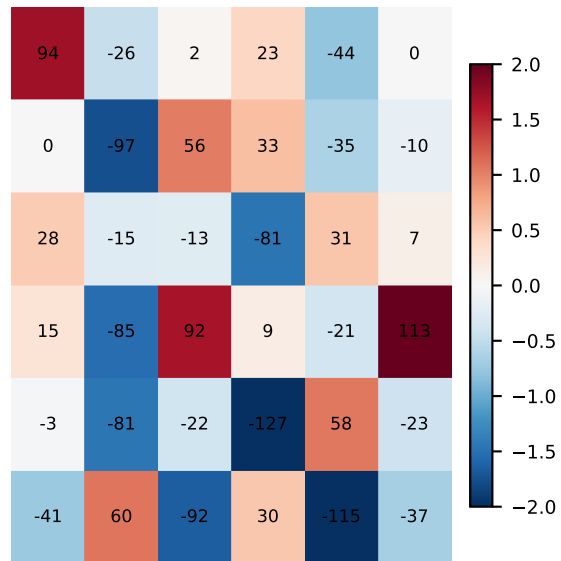


(a) Weight Quantization
(at model load time)

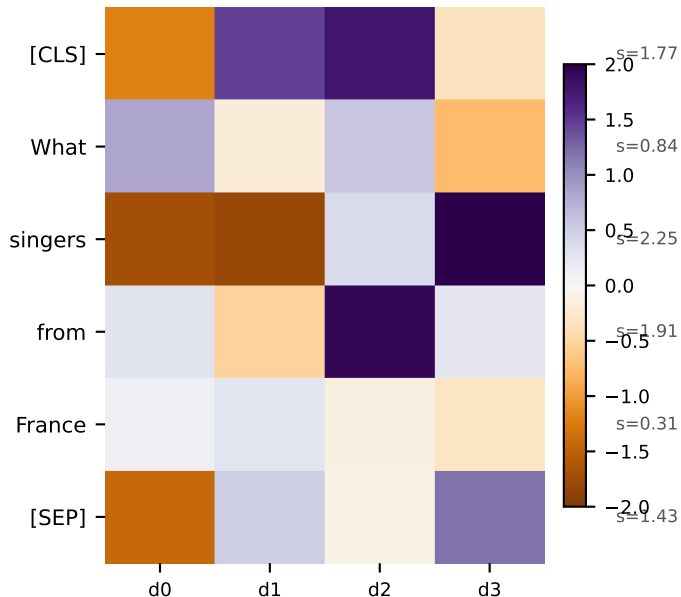


FP32 weights $\rightarrow$ stored as INT8

$$w_{int8} = \text{round}(w_{fp32}/s)$$

$$\text{scale } s = \max(|W|) / 127$$

(b) Dynamic Activation Quant.
(per-token, at inference)



Activation matrix (6 tokens $\times$ 4 dims)

$$a_{int8}[t] = \text{round}(a_{fp32}[t]/s_t)$$

s_t computed fresh for each token t

(c) Effect on Memory & Latency
(arctic-embed-m, CPU ARM64)

	FP32	INT8	Δ
RAM (MB)	418	91	-78%
ms/query	64.1	54.4	-15%
R@5	0.9952	0.9952	0.000
MRR	0.9513	0.9513	0.000
Bits/param	32	8	-75%

**No quality degradation
at 4 $\times$ smaller model**